

Local Setup Instructions

Service Mobility Dashboard

Full-Stack Final Project

This document explains how to run the Service Mobility Dashboard locally from a clean clone of the GitHub repository.

1. Prerequisites

Before running the application, make sure the machine has:

- Git.
- Node.js and npm.
- Access to the shared Supabase project or the required Supabase project values.
- A valid Mapbox public token.
- A Google Maps / Places API key configured for the application.

2. Clone or Update the Repository

For a first-time setup:

```
git clone https://github.com/adigreiman1/fullstack_final_project.git
cd fullstack_final_project
```

If the repository is already cloned, update the local copy before starting:

```
git pull
```

3. Install Dependencies

```
npm install
```

4. Configure Environment Variables

Create a file named `.env.local` in the project root, next to `package.json`:

```
touch .env.local
```

Add the following required application variables:

```
NEXT_PUBLIC_SUPABASE_URL=<your-supabase-project-url>
NEXT_PUBLIC_SUPABASE_ANON_KEY=<your-supabase-anon-key>
NEXT_PUBLIC_MAPBOX_TOKEN=<your-mapbox-public-token>
NEXT_PUBLIC_GOOGLE_MAPS_API_KEY=<your-google-maps-api-key>
```

Security note: do not commit `.env.local` to Git. The repository ignores `.env*` files.

5. Start the Development Server

```
npm run dev
```

After Next.js starts successfully, open:

```
http://localhost:3000
```

If port 3000 is already in use, Next.js may report another available port. Use the URL printed in the terminal, or stop the existing development server and restart the command.

6. Authentication

The dashboard is protected by Supabase Authentication. Use an authorized dispatcher account that already exists in the shared Supabase project. Public self-registration is not part of the application.

Test user credentials and the required environment-variable values are provided separately to the course staff and are not committed to the repository.

7. Run Project Checks and Tests

Lint check:

```
npm run lint
```

Unit tests (Vitest):

```
npm test
```

Unit tests in watch mode:

```
npm run test:watch
```

End-to-End tests (Playwright):

```
npm run playwright install
npm run test:e2e
```

8. End-to-End Test Variables

The Playwright flows use an authorized dispatcher account and controlled test data. Add these values to .env.local only when E2E testing is required:

```
E2E_DISPATCHER_EMAIL=<test-dispatcher-email>
E2E_DISPATCHER_PASSWORD=<test-dispatcher-password>
E2E_SEARCH_ADDRESS=דיזנגוף 10, תל אביב
```

The dispatcher email and password must remain private and must not be committed to the repository. The configured E2E search address is non-secret test configuration.

9. Prepare Deterministic E2E Test Data

Before running `npm run test:e2e`, open the shared Supabase project, go to SQL Editor, and execute `supabase/seed-e2e.sql` from the repository. The script refreshes one dedicated recommendation fixture for the following day at the configured search location. It is test setup

data, not a migration, and it is executed by a privileged database operator rather than by normal application users.

10. Production Build (Optional Local Verification)

Create the production build:

```
npm run build
```

Start the production build locally:

```
npm start
```

11. Troubleshooting

Missing environment variable

If the application reports a missing `NEXT_PUBLIC_*` environment variable, verify that `.env.local` is in the project root, contains the required value, and restart the development server. Next.js reads these variables when the server starts.

Port 3000 is already in use

A previous Next.js process may still be running. Open the existing server if it is the correct project, or stop the process and run `npm run dev` again.

Address search does not work

Verify the Google Maps / Places API key, confirm that the required Google Places API is enabled, and ensure that key restrictions permit local development when restrictions are configured.

Map or route optimization does not load

Verify that `NEXT_PUBLIC_MAPBOX_TOKEN` exists and is valid.

12. Required Environment Variable Summary

Application runtime variables:

- `NEXT_PUBLIC_SUPABASE_URL` - Supabase project URL.
- `NEXT_PUBLIC_SUPABASE_ANON_KEY` - Supabase public anon key.
- `NEXT_PUBLIC_MAPBOX_TOKEN` - Map rendering and route optimization.
- `NEXT_PUBLIC_GOOGLE_MAPS_API_KEY` - Address search with Google Places.

Additional E2E-only variables:

- `E2E_DISPATCHER_EMAIL` - test dispatcher account email.
- `E2E_DISPATCHER_PASSWORD` - test dispatcher account password.

- E2E_SEARCH_ADDRESS - fixed non-secret address used by the E2E-02 decision-support flow (דיזנגוף 10, תל אביב).

Important: never commit real credentials, Supabase project values, or API keys to Git. Keep sensitive values in .env.local or in deployment-platform environment settings. The fixed E2E search address is non-secret test configuration.

Demo data note

For a complete demonstration of the system with populated data, it is recommended to load the demo week 01/09/2026-07/09/2026 using the Demo button on the main dashboard. This provides a consistent sample dataset for reviewing the main product flows during evaluation.